

The empirical recurrence for OEIS A236251: recovering a conjecture that is not written down

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Abstract

OEIS A236251 records “Empirical recurrence of order 48”, with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on a finite digraph and satisfies some monic recurrence of order at most the number of its states with distinct futures, here $S' = 256$; Berlekamp–Massey applied to $2S'$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 48, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 48 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry’s recurrence is the one displayed here, whatever its file contains, and it is proved.

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1 A conjecture one cannot read

OEIS A236251 is “Number of $(n+1) \times (1+1)$ 0..3 arrays colored with the maximum plus the upper median plus the lower median of every 2×2 subblock.”. It has offset 1 and begins

256, 3208, 40120, 505708, 6373768, 80221804, 1009767552, 12713539468, ...

The entry states:

Empirical recurrence of order 48 (see link above)

As of the “Last modified” line on the live entry (Jul 23 2025 09:11:20, revision 8) this is still recorded as empirical. The recurrence’s coefficients are not in the entry text.

2 The count is a walk count

The entry’s condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u , and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 256$ vertices, $a(n) = \iota^\top M^{j(n)} \tau$ for a linear reindexing j fixed by the entry’s shape and checked against its published terms. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \dots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 256$.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S = 512$ exact terms of the model returns order 48 — the order the entry states — with the integer coefficients

$$a(n) = \sum_{i=1}^{48} c_i a(n-i),$$

c_1	0	c_{13}	−454480468	c_{25}	−6836762201243840	c_{37}	−1294990554891550720
c_2	0	c_{14}	30032164944	c_{26}	−8759409963771520	c_{38}	−546168790711009280
c_3	648	c_{15}	146956703544	c_{27}	7753322470999296	c_{39}	4995302197192818688
c_4	8588	c_{16}	275553530575	c_{28}	58955369738388224	c_{40}	5847986080031178752
c_5	61956	c_{17}	−127098485012	c_{29}	143390617243062272	c_{41}	−3454522909857415168
c_6	413472	c_{18}	−4157472558888	c_{30}	−79655839461775360	c_{42}	−5260080579017179136
c_7	1618600	c_{19}	−16080822776528	c_{31}	−852154519853305856	c_{43}	−811648764904210432
c_8	2985994	c_{20}	−12132998766584	c_{32}	−437136524533874688	c_{44}	−114133195619827712
c_9	−8807068	c_{21}	70656575378176	c_{33}	2117718546837151744	c_{45}	1591624484392009728
c_{10}	−152834984	c_{22}	328798681828032	c_{34}	1990656124107014144	c_{46}	1891602448515072000
c_{11}	−731801304	c_{23}	721049343054848	c_{35}	−2093883288551718912	c_{47}	−319310675060981760
c_{12}	−1777355964	c_{24}	−780825278839184	c_{36}	−2821335050330570752	c_{48}	−493965391303802880

Theorem 3. *The recurrence above holds for every n , and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 48, possibly only from some index on. Let q be that recurrence’s characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $48 + 4$ terms removed returns order 48 as well, so $\deg q_{\min} = 48 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. $\square$

4 Verification

Three checks.

First, the walk model reproduces all 15 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once more on the sequence with its first $48 + 4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture’s statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 48 that this sequence satisfies — and that family turns out to have exactly one member.

References

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